Chapter Review

Key Terms

absolute value The absolute value of a number is its distance from 0 on the number line.
integers Integers are counting numbers, their opposites, and zero ... –3, –2, –1, 0, 1, 2, 3 ...
negative number A negative number is less than zero.
opposites The opposite of a number is the number that is the same distance from zero on the number line, but on the opposite side of zero.

Key Concepts

3.1 Introduction to Integers

- Opposite Notation
  - $-a$ means the opposite of the number a
  - The notation $-a$ is read *the opposite of* a.
- Absolute Value Notation
  - The absolute value of a number n is written as $|n|$.
  - $|n| \geq 0$ for all numbers.

3.2 Add Integers

- **Addition of Positive and Negative Integers**

$5+3$	$-5+(-3)$
both positive, sum positive	both negative, sum negative
When the signs are the same, the counters would be all the same color, so add them.	
$-5+3$	$5+(-3)$
different signs, more negatives	different signs, more positives
Sum negative	sum positive
When the signs are different, some counters would make neutral pairs; subtract to see how many are left.	

3.3 Subtract Integers

- **Subtraction of Integers**

$5-3$	$-5-(-3)$
2	-2
2 positives	2 negatives
When there would be enough counters of the color to take away, subtract.	
$-5-3$	$5-(-3)$
-8	8
5 negatives, want to subtract 3 positives	5 positives, want to subtract 3 negatives

Table 3.13

need neutral pairs	need neutral pairs
When there would not be enough of the counters to take away, add neutral pairs.	

Table 3.13

- **Subtraction Property**
  - $a - b = a + (\text{-}b)$
  - $a - (-\text{b}) = a + b$
- **Solve Application Problems**
  - Step 1. Identify what you are asked to find.
  - Step 2. Write a phrase that gives the information to find it.
  - Step 3. Translate the phrase to an expression.
  - Step 4. Simplify the expression.
  - Step 5. Answer the question with a complete sentence.

3.4 Multiply and Divide Integers

- **Multiplication of Signed Numbers**
  - To determine the sign of the product of two signed numbers:

Same Signs	Product
Two positives Two negatives	Positive Positive

Different Signs	Product
Positive • negative Negative • positive	Negative Negative

- **Division of Signed Numbers**
  - To determine the sign of the quotient of two signed numbers:

Same Signs	Quotient
Two positives Two negatives	Positive Positive

Different Signs	Quotient
Positive • negative Negative • Positive	Negative Negative

- **Multiplication by -1**
  - Multiplying a number by -1 gives its opposite: $-1a = -a$
- **Division by -1**
  - Dividing a number by -1 gives its opposite: $a \div (-1) = \text{-}a$

3.5 Solve Equations Using Integers; The Division Property of Equality

- **How to determine whether a number is a solution to an equation.**
  - Step 1. Substitute the number for the variable in the equation.